Franck-Hertz

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1 Abstract

This report investigates the quantization of atomic energy levels through the Franck-Hertz experiment using mercury and neon gas tubes. By measuring the periodic dips in collector current as a function of accelerating voltage, the primary excitation energies were determined. For mercury, the mean peak spacing was found to be 4.99 ± 0.14 V, which is 0.64σ from the literature value of 4.9 eV. For neon, the measured excitation energy of 17.19 ± 0.26 eV differs from the literature value of 16.8 eV by 1.5σ. This suggests that the inelastic collisions were split between the lower 16.6 to 16.8 eV energy levels, as well as the higher 18.3 to 18.9 eV energy levels. This report also analyses how increasing oven temperature raises vapor pressure and gas density, leading to more frequent scattering that shifts the observed current peaks to higher potentials.

2 Theory and Introduction

In 1913, Neils Bohr postulated that bound electrons could only exist in quantised, discrete energy states. A year later in 1914, James Franck and Gustav Hertz published their results after bombarding mercury atoms with electrons. This provided direct evidence for Bohr's quantised energy levels.

The experimental apparatus consists of a vacuum tube filled with a specific gas, such as mercury or neon. A heated cathode produces a beam of free electrons via thermionic emission. These electrons are then accelerated through the gas by a positive potential difference, which pulls the electrons away from the cathode and towards a collecting plate.

As these electrons travel through the tube, they inevitably collide with the gas atoms. To understand the mechanics of these collisions, we must first calculate the respective velocities of the particles. The speed of an electron (v_e) accelerated by a potential difference of just 1 V is (keep in mind that a maximum accelerating voltage of ≈ 100 V will be used in this experiment):

$$v_e = \sqrt{\frac{2eV}{m}} \approx 5.9 \times 10^5 \text{ m/s} \tag{1}$$

Comparing this to the root-mean-square speed (v_A) of the atoms, with M being the mass of an atom[1]:

$$v_A = \sqrt{\frac{3kT}{M}} \tag{2}$$

which gives $v_A \approx 242$ m/s for Mercury at 200 °C and $v_A \approx 607$ m/s for Neon at 25 °C.

We are now ready to discuss two fundamental assumptions made in this experiment.

1. **The atoms are effectively stationary when compared to the electrons.** Because the velocity of the electron ($v_e \approx 10^5$ to 10^6 m/s) is orders of magnitude greater than the thermal velocities of both mercury and neon ($v_A \approx 10^2$ m/s), the gas atoms can be approximated as stationary targets relative to the incoming electron. This ensures that the energy transferred from the gas atoms to the electrons is negligible, thus the collision energy is dictated entirely by the accelerating voltage.

2. **The atoms are much more massive than the electrons.** The mass of the target gas atom (m_2) is five to six orders of magnitude greater than the mass of the incident electron (m_1). An electron has a mass on the order of 10^{-31} kg, while mercury and neon atoms have masses on the order of 10^{-25} kg and 10^{-26} kg, respectively. When a light particle undergoes an elastic collision with a massive, stationary target, the final velocity of the atom (v_{2f}) is approximately[1]:

$$v_{2f} \approx \left(\frac{2m_1}{m_2}\right) v_{1i} \tag{3}$$

   Because $m_2 \gg m_1$, $\frac{2m_1}{m_2} \to 0$, thus the final velocity of the atom is approximately zero. Since kinetic energy is proportional to the square of velocity, the kinetic energy transferred to the massive atom is also effectively zero. Thus, an electron undergoing an elastic collision will bounce off the heavy atom with its momentum redirected but its kinetic energy virtually unchanged.

If all collisions were elastic, this would look like a strictly increasing graph when plotting the collector current against accelerating voltage. However, empirical observations suggest otherwise, that there are periodic dips in the observed collector current. This implies that there are certain energy levels for which the electron collides inelastically, losing energy in the process.

To successfully observe these inelastic collisions, we must filter out all the electrons that have lost their energy. Without this, all electrons would eventually make it to the collector plate, regardless of whether they underwent elastic or inelastic collisions, which would render the current measurements meaningless. This filtering is achieved by applying a small positive retarding voltage (U_2) just before the collector plate, which prevents any electrons lacking sufficient kinetic energy from passing through. Consequently, the measured ammeter current serves as a direct measurement of only the electrons that have retained enough kinetic energy to overcome the barrier and reach the plate.

The objectives of this experiment are to:

1. Record the counter current strength in a Franck-Hertz tube as a function of anode voltage
2. Prove Bohr's postulate of discrete energy levels for electrons in an atom,
3. Calculate Excitation energy E_a between ground and excited states of electrons in both mercury and Ne atoms.

3 Experimental Setup

The experiment was conducted with the following setup:

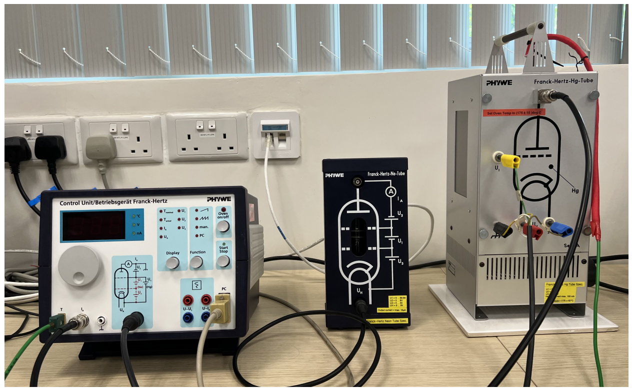

Figure 1: The experimental setup. The Control Unit (Left), Neon Tube (Center) and Mercury Tube (Right).

The overall experimental apparatus, as shown in Figure 1, consists of a central PHYWE Control Unit connected to either the neon or mercury tube enclosures. Because the temperature requirements and internal configurations differ slightly between the two gases, the specific circuit diagrams, component descriptions, and operating temperatures for each element are detailed in their respective sections below.

Data acquisition was automated using the central Control Unit interfaced with a computer. Experimental parameters, such as the accelerating voltage, were configured and executed using the PHYWE “Measure” software. This software simultaneously controlled the accelerating voltage (independent variable) and detected the corresponding collector current (dependent variable), allowing the data to be displayed in real time and exported for analysis.

4 Experiment 1: Mercury

Mercury is an ideal candidate for this experiment for two reasons. Firstly, it has a large atomic mass ($\approx$ 200u) which allows us to preserve our second assumption in the theory section. Secondly, it vaporises into a monoatomic gas of sufficient pressure at temperatures easily achievable in a laboratory (150 to 200 °C in this case).

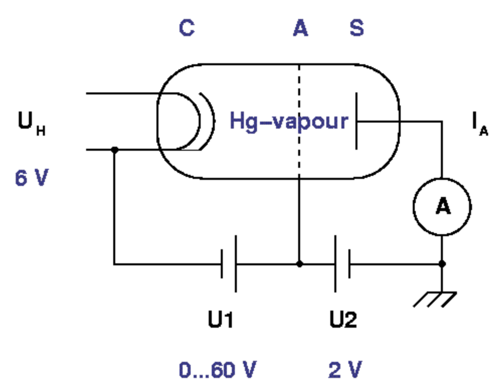

Figure 2: Internal Circuitry of Mercury Franck-Hertz Tube

A heating voltage ($U_H = 6$ V) is applied to the cathode (C) to generate a steady beam of free electrons via thermionic emission. These electrons are accelerated towards a grid-like anode (A) by a variable accelerating voltage (U_1), which is swept from 0 to 60 V. Between the anode and the collector plate (S), an adjustable retarding voltage (U_2) acts as a potential barrier. Only electrons that retain a kinetic energy greater than eU_2 after transiting the tube can overcome this barrier and be recorded as the collector current (I_A) by the ammeter. Finally, to achieve the necessary vapor pressure for these collisions to occur, the entire tube is housed inside a heated oven to vaporise the liquid mercury.

4.1 Vary Retarding Voltage, Keep Temperature Constant

4.1.1 Theory

The purpose of this version of the experiment is to show that the excitation energy is independent of the retarding voltage as well as to show the effects of an increasing retarding voltage on the current. For this procedure, the oven temperature is held at 175°C throughout to maintain a constant mercury vapor pressure.

The retarding voltage (U_2) is systematically increased from 2 V to 4 V while keeping all other parameters fixed. This adjustment changes the potential barrier that electrons must overcome to reach the collector plate, allowing us to observe how the filter threshold affects the measured current (I_A).

4.1.2 Results and Analysis

The results of this experiment have been plotted in the graph below.

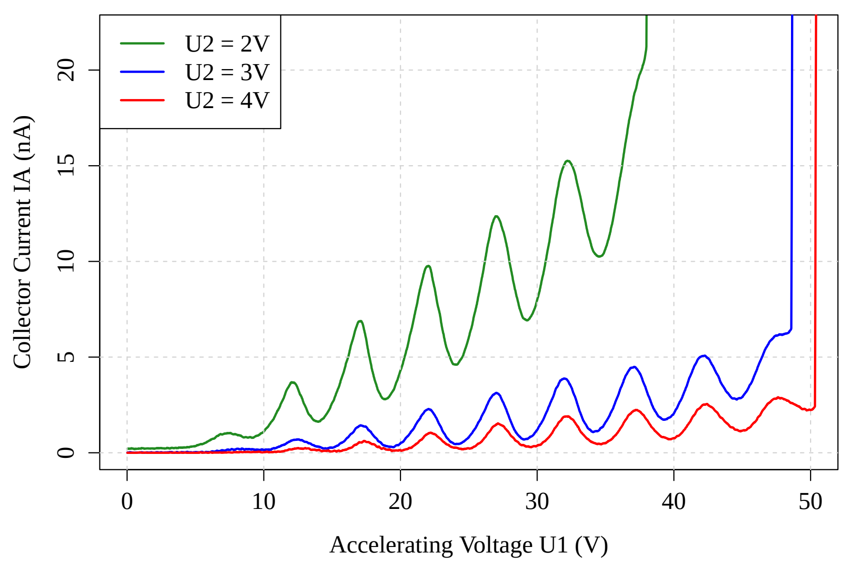

Figure 3: Collector current (I_A) as a function of accelerating voltage (U_1) in a mercury Franck-Hertz tube at 175°C, measured across multiple retarding voltages (U_2).

To accurately determine the excitation energy (E_a), a local least squares quadratic polynomial ($y = ax^2 + bx + c$) was fitted to the data surrounding each peak. The exact voltage coordinate was then calculated using ($x = -b/(2a)$). The voltage differences (ΔV) between adjacent peaks were calculated for each run and averaged across all valid datasets to find the overall experimental excitation energy. This same numerical approach is used to analyze the data for all subsequent experiments.

Retarding Voltage U_2	Mean Peak Spacing ΔV
2 V	5.0 ± 0.3 V
3 V	4.96 ± 0.25 V
4 V	5.0 ± 0.4 V

Table 1: Mean peak spacing and uncertainties for varying retarding voltages at a constant mercury vapor temperature of 175°C. Uncertainties are calculated using the $\frac{\text{max-min}}{2}$ method.

Observations and Discussion:

1. Collector current is inversely related to retarding voltage. A lower retarding voltage means that the electrons have a lower energy barrier to overcome, thus more of the electrons will reach the collector, increasing the reading.
2. Excitation energy is independent of retarding voltage. There are no statistically significant differences (within 1σ) between the average peak spacing (and by extension, excitation energy) of the 3 runs.
3. A higher retarding voltage causes tube ignition to occur at a higher accelerating voltage. At higher accelerating voltages, the bombarding electrons can gain enough energy to ionise the atom, vastly increasing the number of charged particles in the tube and saturating the collector ammeter. This is represented by the vertical lines towards the ends of the graph. Higher retarding voltages can better repel this surge of electrons, thus the 4 V experiment experienced the latest tube ignition.

4.2 Vary Temperature, Keep Retarding Voltage Constant

4.2.1 Theory

The purpose of this version of the experiment is to show that the excitation energy is independent of the temperature, as well as to show the effects of an increasing vapor pressure on the current. For this procedure, the retarding voltage (U_2) is held constant at 1 V throughout to maintain a consistent potential barrier at the collector plate.

The oven temperature is systematically increased from 150°C to 200°C while keeping all other parameters fixed. This adjustment directly changes the vapor pressure and the mean free path λ (average distance that an electron moves before an inelastic collision) of the electrons inside the tube, allowing us to observe how collision frequency affects the measured current (I_A).

4.2.2 Results and Analysis

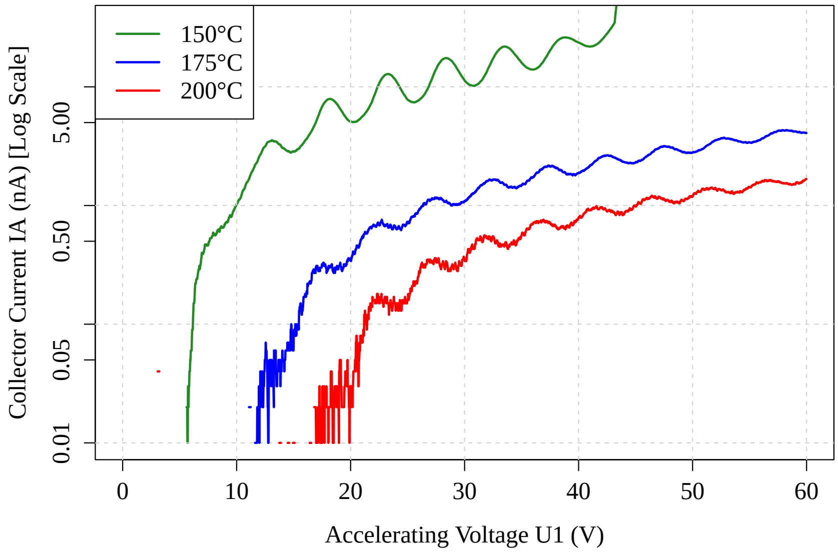

Figure 4: Collector current (I_A) as a function of accelerating voltage (U_1) for a mercury Franck-Hertz tube at varying oven temperatures, plotted with a logarithmic y-axis. The retarding voltage (U_2) is held constant at 1 V.

Oven Temperature	**Mean Peak Spacing (ΔV)**
150 °C	5.14 ± 0.22 V
175 °C	5.03 ± 0.27 V
200 °C	4.86 ± 0.23 V

Table 2: Mean peak spacing and uncertainties for varying oven temperatures in the mercury Franck-Hertz tube. Uncertainties are calculated using the $\frac{\text{max-min}}{2}$ method.

Observations and Discussion:

1. Higher temperatures led to lower collector current. Increasing oven temperatures causes more mercury to vaporise, decreasing the mean free path. This causes more frequent collisions that randomise the electron's forward velocity, so they are less able to penetrate the gas to reach the collector[2].

   This also leads to the peaks occuring "later", as a higher accelerating voltage is needed to overcome the increased scattering and ensure that electrons gain sufficient forward momentum to reach the collector.

2. Increasing temperatures do not affect mean peak spacing. The 3 different experiments give results that are within 1σ of each other.

5 Experiment 2: Neon

The purpose of this version of the experiment is to demonstrate that quantized energy absorption is not unique to mercury by observing the phenomenon in neon gas. This procedure also aims to show that the mean peak spacing, and by extension the excitation energy, is independent of both the control voltage (U_3) and the retarding voltage (U_2).

Unlike the mercury apparatus, the neon Franck-Hertz tube includes an additional control grid to stabilize electron emission. To analyze the system's dependence on these parameters, the control voltage (U_3) is systematically varied while keeping all other settings fixed. Subsequently, the retarding voltage (U_2) is varied independently while holding all other parameters constant, allowing us to observe how changes to the emission and filter thresholds affect the measured collector current (I_A).

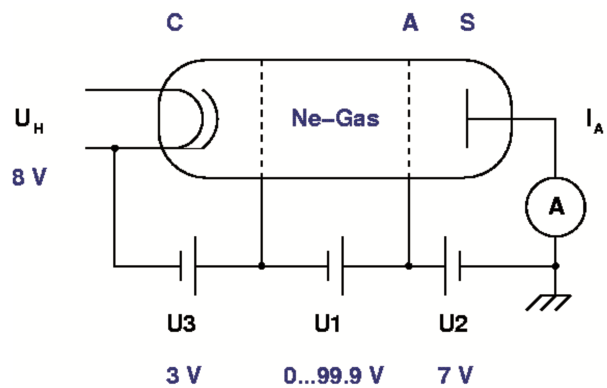

Figure 5: Internal Circuitry of the Neon Franck-Hertz Tube

5.1 Vary Retarding Voltage, Keep Control Voltage Constant

5.1.1 Theory

For the first Neon experiment, the control voltage (U_3) is held constant while the retarding voltage (U_2) is varied. Increasing the retarding voltage should theoretically increase the energy barrier that electrons need to overcome to reach the collector, thus decreasing the overall observed current.

5.1.2 Results and Analysis

The retarding voltage was varied from 5 V to 11 V in intervals of 1 V, but only 3 of the runs are plotted here to increase the clarity of the graph. The rest of the runs followed the trend of the plotted graphs, with no anomalies.

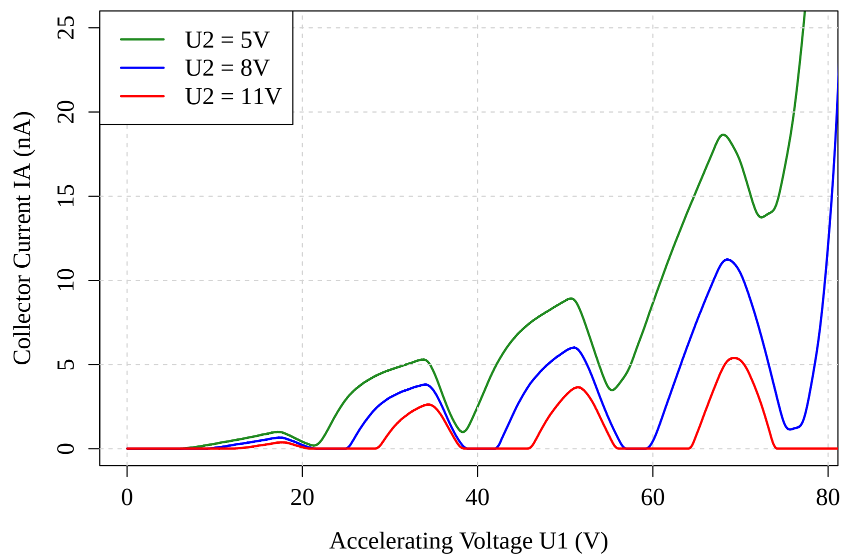

Figure 6: Collector current (I_A) as a function of accelerating voltage (U_1) for a neon Franck-Hertz tube at varying retarding voltages (U_2). The control voltage (U_3) is held constant at 3 V.

Observations and Discussion: The mean peak separation was 17.19 ± 0.11 V (Uncertainties are calculated using the $\frac{\text{max-min}}{2}$ method). Just like mercury,

1. each run returned a value within 1σ of the rest of the data, suggesting that mean peak separation (and thus excitation energy) is independent of retarding voltage;
2. higher retarding voltages led to lower overall current as fewer electrons possessed the necessary energy to overcome the energy barrier.

5.2 Vary Control Voltage, Keep Retarding Voltage Constant

5.2.1 Theory

For the last experiment, the control voltage (U_3) was varied from 2 to 4 V while the retarding voltage was held constant at 9 V. The primary objective is to demonstrate that while the magnitude of U_3 significantly influences the number of electrons extracted from the cathode, it does not alter the fundamental energy levels of the neon atom, and therefore has no effect on the mean peak spacing.

5.2.2 Results and Analysis

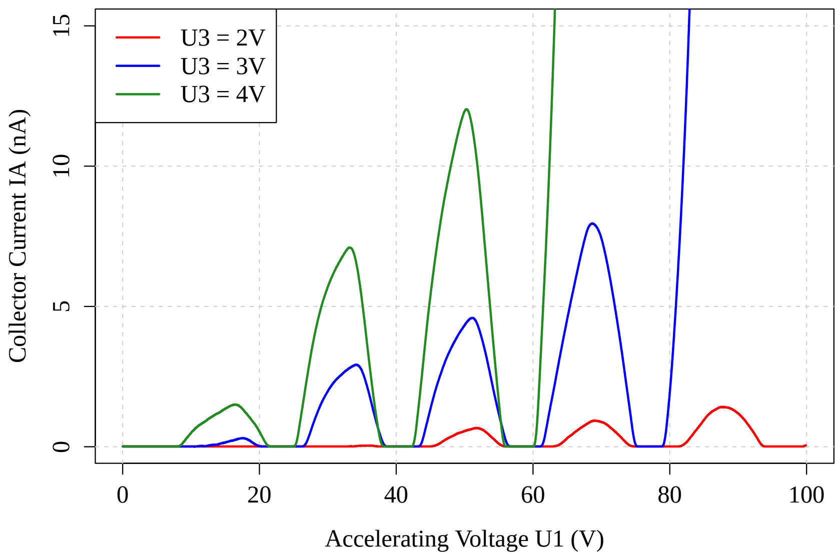

Figure 7: Collector current (I_A) as a function of accelerating voltage (U_1) for a neon Franck-Hertz tube at varying control voltages (U_3). Retarding voltage is held constant at 9 V.

Control Voltage (U_3)	**Mean Peak Spacing (ΔV)**
2 V	17.4 ± 1.5 V
3 V	17.2 ± 0.8 V
4 V	16.9 ± 0.4 V

Table 3: Mean peak spacing and uncertainties for varying control voltages in the neon Franck-Hertz tube. Uncertainties are calculated using the $\frac{\text{max-min}}{2}$ method.

Observations and Discussion:

1. Increasing control voltage leads to higher observed currents. This is because a higher voltage allows the control grid to pull out more electrons from the space around the cathode, increasing the number of electrons that can be accelerated to reach the collector.

2. Mean peak spacing is independent of control voltage, as the 3 experimental runs differed by less than 1σ.

6 Global Analysis

The results from combining the data from all the experimental runs are as follows:

Element	**Mean Peak Spacing ΔV**
Mercury	4.99±0.14 V
Neon	17.19±0.26 V

Table 4: Global mean peak spacing for all mecury and neon experiments.

Note: This model is based on the assumption that the peak spacings are perfectly uniform and purely based on the characteristics of the atom. This is slightly oversimplified, but I will mention more about this in Section 7.5.

We can now compare these results to those of literature. Since the mean peak spacing corresponds to energy changes for a single electron, the excitation energy between ground and excited states have the same numerical value but different units, ie:

$$\text{Excitation energy for mercury, } E_{a,Hg} = 4.99 \pm 0.14 \text{ eV} \tag{4}$$

$$\text{Excitation energy for neon, } E_{a,Ne} = 17.19 \pm 0.26 \text{ eV} \tag{5}$$

Literature values are 4.9 eV for mercury and 16.8 eV for neon. This corresponds to a statistical deviation from literature of 0.64σ for mercury, and 1.5σ for neon. Possible reasons for these deviations will be discussed in the next section.

7 Discussion

7.1 Confirmation of Discrete Energy Levels

This experiment validates Bohr's postulate for discrete energy levels for electrons in an atom. The dips in the graph are the signature of the bombarding electrons undergoing inelastic collisions, only at specific energy levels. The graphs also all show regular peaks, and this feature physically represents the situations were an electron has enough energy to undergo 2,3 or more collisions.

7.2 Comparison with Literature

While both statistical deviations are well within experimental error (0.64σ for mercury, and 1.5σ for neon), the value for neon is slightly higher than expected. This difference can be attributed to the fact that neon has another higher energy level ($\approx$ 18.3 to 18.9 eV). After an electron gains enough energy to undergo an inelastic collision, it still has to travel some distance before it hits an atom. During this time, the electron will gain even more energy, and just a slight increase of 1.5 eV will give it sufficient energy to cause an electron to jump to this $\approx$ 18.5 eV energy level. Thus, the resulting current we observe is a weighted average of the electrons that lose their energy to the 16.8 eV and 18.5 eV energy gaps.

This is further supported by the fact that the peaks in the neon graph, specifically Figure 6, are much less well-defined than the peaks in the mercury graphs. The peaks are instead raised over a range of accelerating potentials, suggesting that there are multiple energy gaps that are contributing to the peaks in the neon graph.

7.3 Sources of Error

7.3.1 Systematic Errors

1. **Mean Free Path:** In an ideal world, the electrons would immediately collide with an atom after gaining sufficient energy. In practice, the electrons have to travel a slight distance after gaining sufficient energy before coming into contact with an atom. This increases the mean peak spacing as electrons gain enough energy to excite atoms to higher energy levels, and also causes peaks to be more smeared out as there is a larger range of energies that an electron can have before colliding with an atom.
2. **Elastic scattering:** The current model only accounts for the energies of the electrons, and not the directions of the electron momenta. While energy theoretically remains constant during elastic collisions, an increase in the number of elastic collisions (especially at high temperatures) randomises the direction of the electrons. This decreases the forward momentum component needed to reach the collector, reducing observed current and requiring higher accelerating voltages to maintain a detectable signal. It is important to note that this effect systematically shifts peaks rightwards, but does not affect the calculation of excitation energy as that is only affected by peak separation and not the absolute accelerating voltage.

7.3.2 Random Errors

1. **Electron Velocity Distribution:** Electrons are emitted from a hot cathode with a range of initial kinetic energies. This "thermal smearing" makes the current minima and maxima broader and less well-defined.
2. **Temperature Fluctuations:** Although temperature was shown to not affect mean peak spacing, it was observed to shift the peaks rightwards for higher temperatures. Therefore, any random fluctuations in temperature (up to 1 °C in this experiment) could have shifted peaks left and right during a single run, leading to random errors in the mean peak spacing.

7.4 Impact of Changing Temperature

The oven temperature can change the thermal energy of the atoms. However, this is not a significant effect. As calculated in Section 2, even at the maximum temperature used during this experiment, the kinetic energy of the electrons were still ~ 3 to 4 orders of magnitude greater than the thermal energy of the atoms. Changing oven temperatures can also change the vapor pressure and density of the mercury gas. This has a much larger impact. At low pressures:

1. The average distance between collisions (mean free path) is larger, so electrons have more time to gain kinetic energy, which could possibly excite electrons in the atoms to higher energy states. This is what was observed in the neon graphs, where the peaks appeared to be more spread out.
2. The fewer collisions also means that the momenta of the electrons are mostly pointing towards the collector plate, increasing the observed current.

At high pressures:

1. The mean free path decreases, and the increased frequency of collisions randomises the directions of the momenta of the electrons. Instead of pointing generally towards the collector, they now point in all directions, even backwards away from the collector. This scattering means that fewer electrons make it to the collector, decreasing observed current.
2. It also means that peaks appear at higher accelerating voltages, since the electrons require a stronger field to attain the required excitation energy while overcoming the increased frequency of momentum-randomizing elastic collisions.

7.5 Non-uniform Peak Spacings

The most surprising discovery for me was that the peaks were not regularly spaced, but instead systematically increased as the number of peaks increased. A sample of the spacings for the mercury experiment with temperature = 175 °C, retarding voltage = 2 V is shown below:

Gap Index ($n \to n+1$)	**Peak Spacing (V)**
$1 \to 2$	4.68
$2 \to 3$	4.83
$3 \to 4$	4.95
$4 \to 5$	5.00
$5 \to 6$	5.09
$6 \to 7$	5.18

Table 5: Individual peak spacings for mercury, showing a systematic increase as the accelerating voltage rises.

As this trend was consistent with every single experimental run, random error cannot account for these discrepancies. To understand why this occurs, picture an electron when the collector current is at its second minimum. The electron collides inelastically with an atom, loses its energy, then is accelerated again by the electric field as it travels the mean free path to its next target. This causes it to gain energy which it again transfers to a bound electron.

Now consider a free electron when the collector current is at its 7$^{\text{th}}$ or 8$^{\text{th}}$ minimum. The electron collides with a bound electron, transfers its energy and is accelerated by the electric field. However, the accelerating voltage is significantly higher now, while the mean free path remains the same (as it is dictated by pressure), thus the electron will have a higher amount of energy by the time it collides with its next target. This means that it is now able to excite the bound electrons to higher energy states. Thus the higher the gap index, the larger the energy of the electron, and the larger the peak spacing corresponding to bound electrons being excited to higher energy states.

This hypothesis is also supported by research done by Rapior et al.[3]

8 Conclusion

The Franck-Hertz Experiment successfully verified Bohr's model of the atom which postulates discrete energy levels for bound electrons. This was shown through the periodic dips in collector current, which signified the voltages at which the energy of the bombarding electron matched an excitation energy of the atom. Analysis also showed that retarding voltage, control voltage and temperature had no effect on the mean peak spacing between experimental runs, while increased temperature led to lower currents and peaks shifted towards higher accelerating voltages. Increasing retarding voltage and decreasing control voltage resulted in overall lower observed currents. The excitation energies were found to be 4.99±0.14 eV (mercury) and 17.19±0.26 eV (neon), which corresponds to a statistically consistent deviation from literature of 0.64σ and 1.5σ for mercury and neon respectively. Finally, discrepancies were accounted for and possible reasons for non-uniform peak spacings were also discussed.

9 References

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